5.7.3 Cybersecurity and electronic warfare vulnerabilities

As drones become more digitally interconnected, they become increasingly vulnerable to cyber intrusions, data spoofing, and electronic warfare (EW) attacks. Adversaries can exploit weaknesses in communication protocols, AI algorithms, or GPS systems to jam, hijack, or mislead drones. The growing reliance on network-centric operations magnifies the risk, as a single compromised UAV can jeopardize the integrity of an entire swarm or mission network. Developing quantum-secure encryption and resilient AI defense systems remains a top priority, but continues to pose complex technical challenges.

5.7.4 Cost and accessibility issues

The integration of AI autonomy, stealth technology, and advanced sensors drives the cost of fourth-generation drones to levels comparable to manned aircraft. This high production and maintenance cost restricts accessibility for developing nations and limits the scalability of these systems for large fleets. The requirement for specialized infrastructure, ground control systems, and data analysis facilities further increases the overall lifecycle expenditure.

Moreover, the testing and certification process for autonomous drones, particularly in civil airspace, remains slow and highly regulated. Achieving international compliance with aviation safety standards for fully autonomous UAVs adds another layer of complexity to their deployment.

5.7.5 Ethical, legal, and regulatory challenges

The growing autonomy of drones raises ethical and legal concerns, especially regarding lethal autonomous weapon systems (LAWS). Questions surrounding accountability, rules of engagement, and international humanitarian law remain unresolved. Determining who bears